

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 04

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

A $(x - 3)^2 + (y - 3)^2 = 40$

B $(x - 2)^2 + (y - 4)^2 = 20$

C $(x - 3)^2 + (y - 3)^2 = \sqrt{20}$

D $(x - 3)^2 + (y - 3)^2 = 20$

E $(x - 3)^2 + (y - 3)^2 = 80$

F $(x + 3)^2 + (y + 3)^2 = 20$

2 Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

A $x = \log_3 4 - 2$

B $x = \log_3 2$

C $x = \frac{\log_3 4}{4}$

D $x = \log_2 3$

E $x = \frac{1}{2} \log_3 2$

F $x = 2 \log_3 2$

3 The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

A 5

B -5

C -3

D 3

E -7

F 7

- 4** Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

- I** If $m_1m_2 = -1$ then the lines meet at exactly one point.
- II** If $m_1 = m_2$ then the lines do not meet.
- III** If the lines meet on the y-axis then $c_1 = c_2$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 5 The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?
- A The estimate is an overestimate.
 - B The estimate is an underestimate.
 - C The estimate is an underestimate only when n is even.
 - D The estimate is exact.
 - E It cannot be decided without calculating.

6 A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3)$.

II So $(x - 2)(x - 3) > 0$.

III So $x - 2 > 0$ and $x - 3 > 0$.

IV So $x > 3$.

Which of the following is correct?

- A** The first error is on line III: a product is positive when both factors are positive **or** both are negative, so $x < 2$ is missed.
- B** The solution is completely correct.
- C** The first error is on line I.
- D** The first error is on line IV: it should be $x > 2$.
- E** The first error is on line II.

- 7 The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 8** Which of the following claims is FALSE, and therefore can be disproved by a counterexample?
- A** If $a > b$ then $a + c > b + c$, for all real a, b, c .
 - B** If $a > b$ then $a^2 > b^2$, for all real a, b .
 - C** If $a > b > 0$ then $a^2 > b^2$.
 - D** If $a > b$ then $a^3 > b^3$, for all real a, b .
 - E** If $a > b$ then $-a < -b$, for all real a, b .

9 Let x be a real number. Which of the following statements is equivalent to ' $x \geq 2$ **if and only if** $x^3 \geq 8$ '?

A $x^3 \geq 8 \Rightarrow x \geq 2$

B $(x \geq 2 \Rightarrow x^3 \geq 8)$ and $(x^3 \geq 8 \Rightarrow x \geq 2)$

C $(x < 2 \Rightarrow x^3 \geq 8)$ or $(x^3 < 8 \Rightarrow x \geq 2)$

D $x \geq 2 \Rightarrow x^3 \geq 8$

E $x \geq 2$ or $x^3 \geq 8$

- 10 Let S be the statement: **every even number greater than 2 can be written as the sum of two primes**. A single counterexample to S would be
- A an odd number that is not the sum of two primes
 - B an even number greater than 2 that can be written as the sum of two primes in two different ways
 - C two primes whose sum is odd
 - D an even number that is the sum of two primes
 - E an even number greater than 2 that is not the sum of any two primes

- 11** n is a positive integer such that n^2 has exactly 3 positive divisors. Which of the following must be true?
- A** $n = 3$
 - B** n is prime
 - C** n is odd
 - D** n is a perfect square
 - E** n is even

12 Claim: the sum of the first n odd numbers is n^2 .

I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.

II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.

III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.

IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

A The proof is completely correct.

B The first error is on line II: you cannot assume the claim.

C The first error is on line III: the next odd number is $2n - 1$.

D The first error is on line IV: it only shows the claim for $n + 1$.

E The first error is on line I.

13 Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

I f is increasing for all $x > 1$.

II f has a local maximum at $x = -1$.

III $f(x) \geq -2$ for all real x .

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

14 Which of the following statements is/are true for all positive real numbers a and b ?

I $\sqrt{a}\sqrt{b} = \sqrt{ab}$

II $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

III $\sqrt{a^2 + b^2} < a + b$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

15 Which of the following statements is/are true for every function f that has a derivative at every real number?

- I** If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.
- II** If f has a local minimum at $x = a$ then $f'(a) = 0$.
- III** If $f'(x) > 0$ for all x , then $f(3) > f(2)$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 16** A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I** The sum of all 10 terms is negative.
- II** The sequence contains at least one negative term.
- III** The sequence contains at least one positive term.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

17 Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4.

I Suppose n^2 is divisible by 4.

II Then $n^2 = 4k$ for some integer k .

III So $n = 2\sqrt{k}$, and since n is an integer $\sqrt{k}$ is an integer.

IV Hence n is even, so $n = 2m$, and $n^2 = 4m^2$ is divisible by 4, so n is divisible by 4.

Which of the following is true?

A The claim is false ($n = 2$ is a counterexample) and the first invalid step is on line IV.

B The claim and proof are correct.

C The claim is true but the first error is on line III.

D The claim is false and the first error is on line II.

E The claim is false and the first error is on line I.

18 Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

I $p(x)$ has three distinct real roots.

II $(x - 4)$ is a factor of $p(x) - 6$.

III $p(x) > 0$ for all $x > 3$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 19 Consider the statement: **there exists a real number k such that for all real x , $x^2 + kx + 1 > 0$.**
Which of the following is true?
- A The statement is true; for example $k = 0$ works.
 - B The statement is false because $x^2 + kx + 1$ can be negative for large x .
 - C The statement is true; every value of k works.
 - D The statement is false because for every k there is an x with $x^2 + kx + 1 \leq 0$.
 - E The statement is true; for example $k = 2$ works.

20 The integers a, b, c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?

I Exactly one of a, b, c is positive.

II $a^2 + b^2 + c^2$ is even.

III One of a, b, c is divisible by 3.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST